

AURORA: A Multi-Objective Reinforcement Learning Framework for Adaptive Cognitive State Optimization in Virtual Reality

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Abstract—Virtual reality (VR) environments pose significant cognitive and physical challenges as users perform concurrent tasks (e.g., navigation, training, etc.), requiring them to manage physical movement, process complex visual scenes, and maintain cognitive performance. These demands often increase cybersickness, elevate cognitive load, and reduce working memory capabilities, as they are interconnected through shared perceptual and cognitive resources. Although several existing visual techniques (e.g., dynamic field of view (DFOV), depth-of-field blur (DOF), etc.) have been proposed and shown to effectively reduce cybersickness, they focus solely on cybersickness and do not consider other important users' cognitive states, such as cognitive load and working memory, and rely on manual adjustment or fixed presets, e.g., aperture size and blur intensity in the DFOV and the DOF, which can significantly degrade overall task performance and hamper user immersive experience (UIX). To address this limitation, we propose *AURORA*, a multi-objective reinforcement learning (RL) framework that jointly optimizes users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities, thereby enhancing UIX and comfort during VR immersion. Specifically, the framework incorporates a multitask learning (MTL)-enabled deep learning (DL) prediction model that predicts these user cognitive states in real time from streaming eye- and head-tracking data, which are then used to construct a multi-objective reward function for RL-based policy learning. Based on this reward, the RL agent dynamically selects visual techniques from our developed visual technique library (e.g., DFOV, DOF, etc.) and adaptively adjusts their intensity, enabling context-aware policies that optimize overall UIX. Furthermore, a human-in-the-loop (HIL)-enabled fine-tuning method has been applied to personalize the learned policy using in-session comfort ratings and post-session feedback, enabling adaptation to individual tolerance profiles. We evaluate *AURORA* with 47 participants in a VR maze simulation on a consumer-grade VR headset (HTC Vive Pro Eye). Results show that the proposed *AURORA* significantly reduces cybersickness, balances cognitive load, and improves working memory, compared with static techniques, with approximately 89% of participants reporting it as more effective, less demanding, and more immersive. The open-source implementation of *AURORA* is available at <https://github.com/dependable-cps/AURORA>.

Index Terms—Reinforcement learning, virtual reality, cybersickness, cognitive load, working memory, personalization.

I. INTRODUCTION

The widespread adoption of consumer-grade virtual reality (VR) headsets (e.g., Meta Quest, HTC Vive, Apple Vision

Pro, Pico, etc.) has significantly transformed the ways users engage with immersive applications across diverse domains, such as healthcare, education, military training, entertainment, and many others [1]–[3]. Many of these applications require users to perform cognitively demanding tasks, such as navigating complex environments while following auditory instructions [4], [5], tracking objects [4], [6], or making decisions under time pressure [2], [6]. These demands place substantial strain on users' cognitive resources, often leading to cybersickness, increased cognitive load, and reduced working memory capabilities [7], [8]. Among these, cybersickness has been the most widely studied due to its immediate and disruptive symptoms [8]–[10] and remains a significant barrier to broader consumer-grade VR adoption [3], [11]. To address this, a variety of visual techniques have been proposed, including dynamic field of view (DFOV) restriction [12], [13], dynamic Gaussian blur (DGB) [14], depth of field (DOF) blur [15], [16], and rest frames (RF) (e.g., virtual nose [17], peripheral teleportation [18], etc.). Although these techniques can effectively reduce cybersickness, they also alter what users perceive and can therefore influence other users' cognitive states [3], [8], [19] (e.g., increasing cognitive load and reducing working memory capacity). For instance, a restricted field of view (FOV) forces compensatory head movements, which increase cognitive load and impair working memory by consuming additional cognitive resources [19], [20]. Similarly, blur-based visual techniques degrade visual interpretability and elevate cognitive load [21]. At the same time, the effectiveness of these techniques depends not only on which technique is applied and its intensity (e.g., the aperture size in DFOV or the blur intensity in DGB), but also on how they are triggered (manually through human intervention or automatically based on predicted user cognitive states). Let us consider a user navigating a VR maze, aiming to find the exit while performing a coin collection task along the way. When rapid head turns are detected, a predefined triggering system applies DFOV with a fixed narrow aperture to reduce cybersickness. Although this reduces discomfort, the narrowed view hides nearby coins, forces additional head movements to see the surroundings, and makes it harder to remember which paths have already been explored, simultaneously increasing cognitive load and degrading working memory. Despite this interdependence among user cognitive states, existing approaches continue to focus primarily on reducing cybersickness without accounting for concurrent effects on other cognitive states (e.g., cognitive load and working memory), nor do they adjust technique intensity to jointly optimize these states [9], [10], [18], [22]–[24].

Prior work on predicting VR users' cognitive states (e.g., cybersickness, cognitive load, and working memory) has pri-

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marily relied on machine learning (ML) and deep learning (DL) models trained on VR sensor data [4], [25]–[34]. For instance, authors in [30] and [32] used DL-based models to predict multiple users' cognitive states from VR simulation data (e.g., eye-tracking, head-tracking, heart rate (HR), etc.). However, these works trained each model separately, failing to capture shared cognitive dependencies, and remained limited to prediction and analysis without using the predicted states to drive adaptive visual technique selection or intensity adjustment. Recently, Kundu et al. [9], [24] proposed a DL-based Transformer model that predicts only the severity of cybersickness and applies an explanation technique to map to predefined visual techniques to reduce cybersickness. However, their work relies on fixed-intensity presets, optimizes a single cognitive state (e.g., cybersickness) using a single-task prediction model, and does not account for other user cognitive states, making the selected technique less effective. These limitations highlight the need for a framework that jointly predicts cognitive states and continuously adapts both visual technique selection and intensity to optimize them beyond fixed presets and single-state objectives, thereby improving user immersive experience (UIX).

To address this critical gap, we propose *AURORA*, a multi-objective reinforcement learning (RL) framework that jointly optimizes users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities, thereby enhancing UIX and comfort during VR immersion. Specifically, *AURORA* incorporates a multitask learning (MTL)-enabled DL model that predicts user cognitive states in real time, whose predictions shape a multi-objective reward that drives an RL agent to dynamically select visual techniques from our developed visual technique library (e.g., DFOV, DGB, DOF, and RF) and adaptively adjust their intensity, enabling context-aware policies that optimize overall UIX. Furthermore, a human-in-the-loop (HIL) fine-tuning process has been applied that incorporates user feedback to personalize the learned policy to individual tolerance profiles. Our key contributions are summarized as follows:

- We first develop a DL-based Transformer model that uses the MTL technique to jointly predict user cognitive states (e.g., cybersickness severity, cognitive physical load level, cognitive mental load level, and working memory capability levels) from streaming VR eye- and head-tracking data. These predicted user states then serve as real-time, multi-dimensional feedback to shape the RL agent's reward during training. We train and validate the developed DL model using the SOTA open-source multitasking dataset, namely *VRWalking* [4]. Experimental results show that the Transformer model achieves prediction accuracies of 93%, 88%, 94%, and 90% for cybersickness, cognitive physical load, cognitive mental load, and working memory, respectively.
- Next, we develop and train an RL agent in domain-randomized VR maze simulations [35] with diverse variations in layout, motion dynamics, and visual flow, improving the agent's ability to learn policies that adaptively select visual techniques and their intensities based on user real-time cognitive states and motion context. The agent is

trained using the Proximal Policy Optimization (PPO) algorithm [36], where predictions from the MTL-enabled DL model shape a multi-objective reward that guides the agent to select the most suitable visual technique and intensity from our developed visual technique library, which contains *DFOV* with an adjustable aperture size, *DGB* with tunable blur intensity, *DOF* with controllable focal distance, and *RF* (a virtual nose) with adjustable width.

- Finally, we design an HIL-enabled fine-tuning method to personalize the learned policy using in-session comfort ratings and post-session feedback, thereby constructing a user-specific preference vector. This vector is then used to refine the RL reward weights and adjust technique-specific intensity bounds during fine-tuning, enabling the policy to progressively align with each user's tolerance profile.

We validate *AURORA* through a user study with 47 participants in a VR maze simulation on a consumer-grade VR headset (HTC Vive Pro Eye). Our results show that *AURORA* significantly improves key user states during VR immersion by reducing cybersickness, balancing cognitive load, and improving working memory abilities. By adaptively selecting visual techniques and continuously adjusting their intensities, our approach outperforms static and rule-based methods, with approximately 89% of participants reporting superior comfort. *To the best of our knowledge, AURORA is the first framework to formulate VR user cognitive state optimization as a multi-objective RL problem that jointly reduces cybersickness, balances cognitive load, and improves working memory through continuous intensity tuning and HIL personalization. To support reproducibility and provide a foundation for future research, we release AURORA as an open-source¹ framework that includes the VR simulation, RL training pipeline, DL-based prediction model, and HIL personalization module. Our goal is to enable researchers and developers to systematically study and extend adaptive visual technique selection in VR, with the aim of improving user comfort.*

II. BACKGROUND

In this section, we discuss key directions related to our work, including user cognitive states prediction, visual techniques for adaptation, reinforcement learning in VR, and the relationships among these user states and visual techniques.

a) *User Cognitive State Prediction*: User cognitive states (e.g., cybersickness, cognitive load, and working memory) refer to users' internal conditions that shape how they perceive, process, and respond to a VR environment during immersion [4], [30]. These states are closely interrelated and collectively influence user comfort, task performance, and overall UIX [2], [7], [8]. Among them, cybersickness manifests as symptoms such as dizziness, eye strain, and nausea caused by visual-vestibular mismatch [11]. While cognitive load reflects the demands placed on users' limited cognitive resources during task execution [7], and working memory supports the temporary storage and manipulation of task-relevant information [8]. SOTA methods for measuring user cognitive states can

¹<https://github.com/dependable-cps/AURORA>

TABLE I
COMPARISON OF PRIOR WORK WITH OUR PROPOSED AURORA
FRAMEWORK ACROSS USER COGNITIVE STATE (UCS) PREDICTION,
ADAPTATION, AND JOINT OPTIMIZATION OF UCS (JUCS).

Literature Review	UCS Prediction			UCS Adaptation			JUCS
	CS	CL	WM	CS	CL	WM	
Kundu et al. [25]	✓	×	×	×	×	×	×
Jeong et al. [38]	✓	×	×	×	×	×	×
Qi et al. [27]	✓	×	×	×	×	×	×
Zhu et al. [33]	✓	×	×	×	×	×	×
Wei et al. [29]	×	✓	×	×	×	×	×
Gao et al. [31]	×	✓	×	×	×	×	×
Pavel et al. [26]	×	✓	×	×	×	×	×
Setu et al. [32]	×	✓	✓	×	×	×	×
Kundu et al. [30]	✓	✓	✓	×	×	×	×
Fernandes et al. [12]	×	×	×	✓	×	×	×
Hussain et al. [15]	×	×	×	✓	×	×	×
Kelly et al. [13]	×	×	×	✓	×	×	×
Nie et al. [18]	×	×	×	✓	×	×	×
Islam et al. [10]	✓	×	×	✓	×	×	×
Hadadi et al. [34]	✓	×	×	✓	×	×	×
Chiossi et al. [22]	×	×	✓	×	×	✓	×
Matam et al. [23]	×	✓	×	×	✓	×	×
Kundu et al. [9], [24]	✓	×	×	✓	×	×	×
Our proposed	✓	✓	✓	✓	✓	✓	✓

be broadly categorized into three categories: subjective, objective, and ML/DL-based automatic prediction methods. Subjective measures include the Simulator Sickness Questionnaire (SSQ) [37], which assesses cybersickness severity before and after immersion; the Fast Motion Sickness (FMS) scale [28], which measures cybersickness during immersion, the NASA Task Load Index (NASA-TLX) [4], which measures cognitive load across multiple dimensions (e.g., mental, physical, and temporal demand), and the Digit Span Task (DST) [8], which assesses working memory through digit recall. In contrast, objective measures include integrated VR sensor data (e.g., eye- and head-tracking) and physiological signals (e.g., HR and galvanic skin response (GSR)) [4], [28] to assess the user's cognitive state. Building on these modalities, recently, ML and DL methods have been developed to automatically predict user cognitive states from multimodal VR sensor data [4], [25]–[32], [38], as summarized in Table I. For instance, Geo et al. [31] used an ML-based LightGBM model to predict cognitive load from eye-tracking data, while authors in [27] used a DL-based LSTM model to predict cybersickness severity from VR sensor data, yet both focused on a single cognitive state (i.e., cognitive load or cybersickness) separately. Recently, authors in [30] and [32] used DL-based LSTM, GRU, and Transformer models to simultaneously predict multiple users' cognitive states (e.g., cognitive load, working memory). However, they trained separate models for each state prediction and do not jointly optimize these predicted cognitive states to balance cognitive load or reduce cybersickness severity, thereby hindering broader VR adoption. AURORA addresses this limitation through a unified MTL-enabled DL model that jointly predicts multiple cognitive states in real time and uses these predicted states to optimize UX.

b) Visual Techniques for Adaptation: Several visual techniques have been proposed to improve user comfort during immersion [10], [12], [13], [15], [17], [24], [34], which can be broadly classified based on whether they use static or dynamic intensity and whether human intervention (HI) is required for activation. For instance, Fernandes et al. [12] proposed a subtle DFOV adjustment that narrows the FOV during rapid motion, while the author in [15] applied DOF blur to reduce visual conflict. However, these approaches rely on fixed-

intensity presets and often require manual activation, limiting their adaptability to real-time changes in user cognitive states. Beyond fixed, user-triggered approaches, a few studies have introduced automatically triggered visual techniques based on predicted cybersickness [10], [24]. For instance, Islam et al. [10] proposed a closed-loop framework that adjusts visual techniques based on predicted cybersickness severity, while authors in [24] proposed a Transformer-based prediction model that maps predicted features to predefined visual techniques. However, these techniques still apply fixed intensity and focus solely on cybersickness. Although a few works have targeted other cognitive states individually [22], [23], none jointly optimize users' cognitive states across cybersickness, cognitive load, and working memory. As shown in Table I, prior work primarily focuses on a single cognitive state with fixed-intensity presets, without jointly optimizing across these cognitive states. Our work addresses this gap by adaptively selecting visual techniques and continuously adjusting their intensity to jointly optimize users' cognitive states.

c) Reinforcement Learning in VR: RL has been increasingly explored for adaptive and user-centered VR applications due to its ability to learn context-aware policies through real-time feedback [36], [39]–[41]. For example, Lee et al. [41] developed a multi-agent RL system that dynamically switches redirected walking strategies based on the user's environment, while authors in [39] used RL to personalize rehabilitation arm movement trajectories based on patient performance. Similarly, Gehrke et al. [40] demonstrated RL-based adaptive haptic control in XR using both explicit user ratings and neural signals as feedback. These works demonstrate the potential of RL for real-time personalization in VR. However, no prior work has applied RL to jointly optimize users' cognitive states, a gap that AURORA addresses through a multi-objective RL formulation for user cognitive state optimization via adaptive visual technique selection and intensity control.

d) Relationship Among User Cognitive States and Visual Techniques: Users' cognitive states are closely interconnected because they depend on shared and limited perceptual and cognitive resources [7]. Immersive VR further strains these resources by requiring users to process continuous sensory input, maintain spatial orientation, and perform concurrent tasks [2]. This strain manifests across multiple user states, as visually induced motion sickness can degrade short-term memory, reduce multitasking performance, and impair working memory [7], [8], [42]. Furthermore, prior work suggests that the relationships among those cognitive states are bidirectional and context dependent [5], [6], [43]. For instance, moderate cognitive load can reduce cybersickness while excessive workload worsens it [6], and higher secondary-task demands can simultaneously increase cybersickness, elevate cognitive load, and degrade working memory [43]. These interdependencies extend directly to visual adaptation techniques, as each technique interacts with users' multiple cognitive states depending on the context and intensity of its application [8], [17], [19]–[21]. For instance, fixed-intensity DFOV and DGB can reduce cybersickness but may elevate cognitive load and impair working memory due to limited visual information [19]–[21], while a fixed focal distance in DOF may alter depth perception,

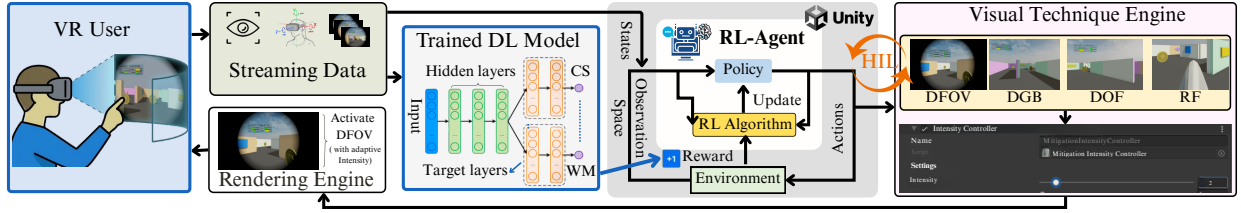


Fig. 1. Overview of the proposed AURORA framework. The MTL-enabled DL model predicts users' targeted cognitive states from streaming VR data, which shapes the RL agent's reward. The RL agent then selects the most suitable visual technique and its intensity via the adaptive visual technique engine, while a HIL fine-tuning process further personalizes the policy.

affecting both cybersickness and working memory [8], and a fixed-width RF may provide limited cybersickness relief in high-motion contexts [17]. Importantly, these effects are intensity-dependent, as a technique that improves one cognitive state at moderate intensity may degrade others at higher intensity [21], [22]. These findings demonstrate that both the choice and intensity of visual techniques must account for their joint effects across these interconnected cognitive states.

III. AURORA FRAMEWORK

This section presents the proposed AURORA framework, that jointly optimizes users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities.

A. System Overview

AURORA operates as a closed-loop system within a VR environment, as illustrated in Figure 1. The framework consists of an MTL-enabled DL model, an RL agent, a Visual Technique Engine, and a HIL personalization process. During VR immersion, the MTL-enabled DL model continuously processes streaming eye- and head-tracking data to predict user cognitive states (e.g., cybersickness severity, cognitive physical load level, cognitive mental load level, and working memory capabilities) in real time. These predictions shape a multi-objective reward signal that guides the RL agent to jointly optimize users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory. At each decision step, the agent receives the predicted user states, along with the currently active visual technique (DFOV, DGB, DOF, and RF) and its intensity (e.g., FOV, blur, etc.), and selects the most suitable technique and intensity for real-time application. Furthermore, user feedback collected during and after each session is combined with the predicted user cognitive states to build a personalized preference vector that drives HIL-based offline policy fine-tuning between sessions.

B. MTL-Enabled DL Model for RL Reward Shaping

For an effective visual technique (e.g., DFOV, DGB, etc.) selection with accurate intensity, we require real-time predictions of the targeted user's cognitive states (e.g., cybersickness, cognitive load, and working memory), since these predictions directly shape the reward used for RL policy learning. Accurate predictions help the RL agent relate its visual-technique actions to their effects on user state, thereby supporting more stable learning and better long-term policy behavior. Therefore, we develop an MTL-enabled DL model that jointly predicts cybersickness severity, cognitive physical

load, cognitive mental load, and working memory levels within a single unified architecture. Specifically, we use a SOTA Transformer-based DL model, which has been increasingly adopted in recent VR user-state prediction studies [24], [26], [30]. The Transformer model includes a positional encoding (PE) layer for input projection and temporal encoding, a Transformer encoder with 64-dimensional multi-head self-attention, residual add-and-norm layers, and a 256-dimensional feed-forward MLP layer (256 dimensions) to capture latent temporal features from VR sensor data, followed by task-specific output layers for severity prediction. Unlike single-task models, which are trained independently for each target state and produce one prediction at a time [30], [32], our proposed MTL-enabled DL model is trained jointly on all target states using a shared backbone with task-specific prediction heads. This shared design is particularly suitable for our RL reward-shaping pipeline because these user states are closely coupled during VR immersion. We train the model on the open-source VRWalking dataset [4] using task-specific categorical cross-entropy losses. Although each task has a different number of class labels, categorical cross-entropy is applied separately to each task's output, and the overall MTL objective is jointly optimized within the shared architecture. For instance, in the cybersickness prediction task, have four severity classes (e.g., none, low, medium, and high), while cognitive load and working memory tasks have three class labels (e.g., low, medium, and high). Thus, the same loss function is used across all tasks, while each task retains its own label space within the shared MTL framework. Because the RL agent requires continuous feedback rather than only discrete class labels, we convert each softmax output into a normalized continuous severity score, as defined in Equation 1 in Section III-C, so that all outputs are mapped to $[0, 1]$ regardless of the number of classes. These normalized severity scores are then used to compute the multi-objective reward in Equation 2. For deployment, we export the trained model to ONNX and integrate it into the Unity runtime.

C. Multi-Objective RL Agent

The RL agent in AURORA is designed to jointly optimize users' cognitive states, which we formalize this process as a Markov Decision Process (MDP), represented by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

State ($s_t \in \mathcal{S}$): The state encodes multimodal features that are informative for the targeted user's cognitive states (e.g., cybersickness (CS), cognitive mental load (CML), cognitive physical load (CPL), and working memory (WM)). Specifically, it includes locomotion signals (e.g., walking speed

and angular velocity), head-tracking signals (e.g., position, rotation, and rotational velocity), eye-tracking signals (e.g., pupil diameter, gaze direction, and fixation stability), visual motion cues (e.g., optic flow magnitude), and task context (e.g., turning vs. straight walking). To capture temporal dependencies, the previously applied visual technique and its intensity are also included.

Action ($a_t \in \mathcal{A}$): At each timestep t , the agent selects a composite action $a_t = (M_t, I_t)$, where $M_t \in \{\text{DFOV, DGB, DOF, RF}\}$ specifies one visual technique from our visual technique library and $I_t \in [0, 1]$ denotes the corresponding normalized intensity. Only one visual technique is applied at each timestep.

Transition dynamics (P): The transition function $P(s_{t+1} | s_t, a_t)$ describes how the user's multimodal state changes in response to different visual-technique actions, including changes in motion patterns, head and eye behavior, visual motion cues, and targeted user states.

Reward (R): The reward is derived from the predictions of the DL-based MTL model (see Section III-B) for the targeted users' cognitive states (e.g., CS, CML, CPL, and WM), unlike prior single-objective formulations that focus only on reducing cybersickness [9], [10], [15], [24]. Let $\hat{s}_{k,t}$ denote the normalized continuous prediction value for user state k at time t , computed as:

$$\hat{s}_{k,t} = \frac{1}{K_k - 1} \sum_{i=0}^{K_k-1} i \cdot p_{k,i,t} \quad (1)$$

where K_k is the number of classes for users' cognitive state k ($K_{\text{CS}} = 4$ and $K_{\text{CPL}} = K_{\text{CML}} = K_{\text{WM}} = 3$), and $p_{k,i,t}$ is the softmax probability of class i at time t from the MTL-enabled DL model in Section III-B. The factor $(K_k - 1)^{-1}$ normalizes all user-state predictions to $[0, 1]$ regardless of class count so that each state contributes on a comparable scale to the reward. The reward at each timestep is then given by:

$$r_t = \sum_{k \in \mathcal{K}} \alpha_k (\hat{s}_{k,t-1} - \hat{s}_{k,t}) - \lambda_m I_t - \lambda_\Delta |I_t - I_{t-1}| - \lambda_{\text{switch}} \cdot \mathbb{I}[M_t \neq M_{t-1}] + \lambda_{\text{stab}} \cdot (1 - |I_t - I_{t-1}|) \quad (2)$$

where $\mathcal{K} = \{\text{CS, CPL, CML, WM}\}$ and α_k is the weight assigned to users' cognitive state k . The penalty terms regulate excessive visual-technique intensity (λ_m), abrupt intensity changes (λ_Δ), and frequent technique switching (λ_{switch}), while the stability bonus (λ_{stab}) encourages smooth transitions. This formulation enables the agent to discover trade-offs across targeted user states. This formulation enables the agent to discover trade-offs between competing user cognitivstates. For example, visual techniques (e.g., DFOV, DGB, etc.) designed to reduce cybersickness may also increase cognitive load and degrade working memory [7], [8], [19]. The agent therefore learns to select intensity levels that balance these user cognitive states rather than optimizing any single state in isolation.

Discount factor (γ): The parameter $\gamma \in [0, 1]$ controls how the RL agent balances immediate and long-term effects of its actions. A moderate value encourages the agent to consider the cumulative impact of visual technique adaptation on user states over time. We train the RL agent using Proximal Policy Optimization (PPO) [36]. We select PPO because it can

TABLE II
PPO HYPERPARAMETERS (HP) FOR AURORA TRAINING.

HP	Value	HP	Value
Learning rate (η)	1.0×10^{-4}	α_{CS}	0.50
Clipping ratio (ϵ)	0.2	α_{CPL}	0.15
Entropy coeff. (β)	1.0×10^{-2}	α_{CML}	0.15
Discount factor (γ)	0.995	α_{WM}	0.10
Mini-batch size	512	λ_m	0.05
Buffer size	10240	λ_Δ	0.10
Num. epochs per update (K)	5	λ_{switch}	0.08
Max training steps	5.0×10^5	λ_{stab}	0.05

effectively handle our mixed action space, which involves both discrete visual-technique selection and continuous intensity control, while maintaining stable policy updates through its clipped surrogate objective, which is critical in our setting where reward signals from the prediction model carry inherent noise. Training is performed in domain-randomized VR maze environments with variations in layout, motion dynamics, visual flow, and walking speed to improve policy robustness. PPO updates are carried out over mini-batches and multiple epochs, guided by the multi-objective reward parameters (α_k , λ_m , λ_Δ , λ_{switch} , λ_{stab}), which are tuned through preliminary experiments, as discussed in section IV.

D. Adaptive Visual Technique Engine

During runtime, the RL policy observes the current cognitive state and outputs an action $a_t = (M_t, I_t)$, where M_t denotes the selected visual technique and I_t denotes its intensity. This action is passed to the adaptive visual technique engine, which includes four SOTA visual techniques: DFOV [12], DGB [14], DOF [15], and RF (virtual nose [17]). As discussed in Section II, these techniques affect user cognitive states differently across motion contexts and individual users, so no single fixed strategy performs well in all VR scenarios. The RL agent, therefore selects the most suitable visual technique and its intensity based on the current context and predicted user cognitive states. Each decision is then evaluated through the multi-objective reward in Equation 2, enabling the policy to jointly optimize users' cognitive states by reducing cybersickness while balancing cognitive load and improving working memory during VR immersion. We implement DFOV using Unity's Tunneling Vignette Controller, where the aperture fraction narrows the visible field from 120° to 60° as intensity increases. DGB is implemented using Unified Universal Blur within Unity's Universal Render Pipeline (URP), where the intensity parameter controls blur strength ($\sigma = 0$ to 15). DOF is implemented through URP's post-processing depth buffer with Bokeh mode, allowing focal distance and blur strength to be adjusted continuously. For RF, following the virtual nose approach [17], we attach a nose model to the user's view and allow the RL agent to adaptively control its width based on the current user cognitive state. Although we focus on these four techniques, the engine can be extended to include other adjustable visual techniques, such as color manipulation [3], peripheral teleportation [18] and snap turning [13], as long as their effects can be modeled within the same RL framework.

E. HIL-Based Personalization of Visual Techniques

The baseline AURORA policy is designed for a general population, but tolerance to visual techniques and preferences

for their intensity can vary substantially across users [3]. For example, some users may tolerate stronger blur but find FOV restriction disruptive, whereas others may prefer minimal visual alteration even if it provides less relief. To account for this variability, we incorporate a HIL personalization process that fine-tunes the policy using user-specific feedback collected during and after immersion. During VR sessions, participants periodically report their current user cognitive state through verbal feedback, while the MTL-enabled DL model concurrently predicts the remaining targeted user cognitive states in real time (Section III-B). Together, these reported and predicted states form a per-technique response profile that captures whether a given visual technique, at its applied intensity, improves the user's overall condition or introduces undesirable side effects. After each session, participants also complete a structured preference questionnaire that captures technique-specific comfort, perceived visual disruption, preferred intensity level, and overall technique ranking. We combine these feedbacks to construct a user-specific preference vector $\mathbf{z}$, which encodes personalized adjustments to the reward weights α_k in Equation 2, technique-specific intensity bounds, and relative preferences over visual techniques. We then use $\mathbf{z}$ to refine the reward parameters and impose technique-specific constraints during offline policy fine-tuning with PPO between sessions, after which the personalized policy is deployed in the next evaluation session. For example, if a user reports that high-intensity DGB is uncomfortable while DFOV remains acceptable, the fine-tuned policy reduces the allowable DGB intensity range and shifts preference toward DFOV, while continuing to optimize across all targeted user states.

IV. DATASETS & EXPERIMENTAL SETUP

This section details the dataset used for training and evaluation, Experimental Setup, and system performance analysis.

Dataset: We use the open-source VRWalking dataset [4], which includes eye-tracking, head-tracking, and physiological data from 39 participants (23 male, 16 female; mean age 25.67, $SD = 7.22$) navigating a virtual maze by real walking for 15 minutes. The dataset provides eye-tracking features (e.g., gaze direction, pupil position, etc), head-tracking features (e.g., position, rotation, etc.), and physiological signals (e.g., HR, GSR) with ground-truth labels for six multitasking metrics: cybersickness, cognitive mental load, cognitive physical load, working memory, attention success rate, and attention reaction time. In this work, we specifically focus on cybersickness, cognitive mental load, cognitive physical load, and working memory to train the DL-based MTL prediction model and shape the RL reward. Following prior studies [9], [25], cybersickness is categorized into four severity levels (none, low, medium, high), while the remaining user cognitive states retain the original three levels (low, medium, and high) as provided in the VRWalking dataset [4], [32]. These discretized severity labels serve as ground truth for training the MTL-enabled DL model (Section III-B). It is worth mentioning that we rely solely on integrated sensor data (e.g., eye and head tracking) rather than external physiological signals (e.g., HR, GSR), as these require tethering to the user's body and are prone to noise [24], [28].

TABLE III

MTL-ENABLED TRANSFORMER MODEL PERFORMANCE FOR DIFFERENT USER COGNITIVE STATES (UCS) ACROSS SEVERITY LEVELS (NONE (N), LOW (L), MEDIUM (M), AND HIGH (H)), WHERE CPL, CML, AND WM HAVE THREE LEVELS (L, M, H) AND A DENOTES ACCURACY.

UCS	A(%)	Precision (%)				Recall (%)				F1-score (%)			
		N	L	M	H	N	L	M	H	N	L	M	H
CS	93	98	90	85	96	96	88	82	95	97	89	83	95
CPL	88	—	90	82	92	—	87	79	90	—	88	80	91
CML	94	—	95	89	96	—	93	87	94	—	94	88	95
WM	90	—	92	84	93	—	89	82	91	—	90	83	92

Experimental Setup: To validate the proposed AURORA framework, we use a VR maze simulation consistent with the environment in the VRWalking dataset [4], ensuring compatibility between the training data distribution and the deployment setting. The system is developed in Unity and rendered using the URP. AURORA is deployed on a consumer-grade head-mounted display (HMD) HTC Vive Pro Eye with integrated eye tracking and head tracking, a resolution of 2880×1600 pixels per eye, and a refresh rate of 90 Hz. The complete system, including RL training and model inference, runs on a workstation equipped with an Intel Core i9-10900X CPU, 128 GB of RAM, and an NVIDIA RTX 3090 GPU.

a) *Virtual Environment and Apparatus:* To ensure compatibility between training and deployment, we replicate a VR maze environment consistent with the VRWalking dataset [4], on which both the MTL-enabled Transformer model and the RL agent are trained. The maze is deployed within a $5\text{ m} \times 5\text{ m}$ physical walking area, with each platform measuring approximately $0.5\text{ m} \times 0.5\text{ m}$. Maze layouts are generated procedurally using a depth-first search-based algorithm, producing connected structures with varied paths across trials. The environment includes colored walls, a cold sunset sky, and no ceiling, as shown in Figure 4a. The HTC Vive Pro Eye streams real-time eye- and head-tracking data to the MTL-enabled Transformer model and the RL policy during execution.

b) *RL Training Setup and Hyperparameter Selection:* The RL agent is implemented using the Unity ML-Agents Toolkit [44] and trained with PPO [36]. The MTL-enabled Transformer model is implemented in PyTorch 2.3. To improve generalization across users and environments, we train the RL agent in domain-randomized maze environments [35] with variations in layout complexity, corridor width, turn frequency, and visual appearance. This exposes the agent to diverse motion and contextual conditions during training. At each decision step, the agent receives a sliding window of recent sensor observations and outputs both a visual technique and its corresponding intensity. The PPO hyperparameters are summarized in Table II. We initialize them based on the original PPO formulation [36] and prior large-scale studies on on-policy RL training [45], and then refine them through grid search over preliminary training runs [46]. During this RL agent training process, we jointly tune the reward weights α_k in Equation 2 and the penalty coefficients (λ_m , λ_Δ , λ_{switch} , λ_{stab}). The final reward configuration assigns the largest weight to cybersickness ($\alpha_{\text{CS}} = 0.50$), followed by cognitive physical load and cognitive mental load ($\alpha_{\text{CPL}} = \alpha_{\text{CML}} = 0.15$), and working memory ($\alpha_{\text{WM}} = 0.10$). This weighting reflects the greater optimization pressure required to reduce cybersickness while balancing cognitive load and improving working mem-

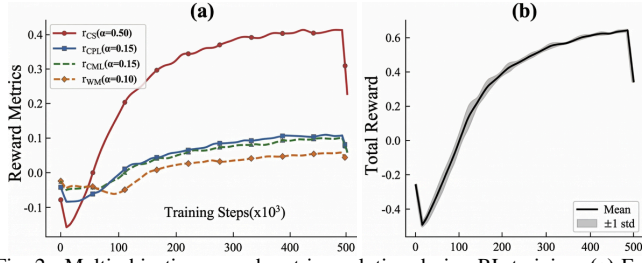


Fig. 2. Multi-objective reward metric evolution during RL training. (a) Each curve represents the contribution of one metric to the total reward. (b) Mean total reward with variance over training steps.

ory. Training is stopped when the cumulative reward on a held-out set of unseen maze configurations no longer improves.

System Benchmarks: For real-time deployment on a consumer-grade VR system, both the DL-based prediction model and the RL policy must execute within the VR frame budget. At 90 Hz, each frame allows approximately 11.1 ms for all processing, including scene rendering, input handling, and AURORA's inference pipeline. In our implementation, the MTL-enabled DL model runs via Unity Sentsis on the GPU in 1.8 ms, the RL policy executes on the CPU Burst backend in 0.3 ms, and the visual technique engine render pass completes in 0.9 ms, yielding a total pipeline overhead of approximately 3.0 ms. This is less than 27% of the per-frame budget, This is less than 27% of the per-frame budget, remaining within acceptable limits for SOTA real-time VR settings [24].

V. TECHNICAL EVALUATION RESULTS

This section presents the technical evaluation of the AURORA framework, including the MTL-enabled DL model for RL reward shaping, the reward metric evolution during RL training, and the learned action distribution analysis.

MTL-enabled DL Model for RL Reward Shaping: Before integrating the MTL-enabled DL model into RL training, we evaluate its performance in terms of predictive accuracy, model size, training time, and inference time. The model has a total size of approximately 410,000 KB and requires roughly 380 minutes to train on the workstation described in Section IV. Table III summarizes the accuracy, precision, recall, and F1-score across all four targeted user cognitive states (e.g., cybersickness, cognitive mental load, cognitive physical load, and working memory). Overall, the model achieves consistently strong performance across all cognitive states, with precision, recall, and F1-score remaining generally high throughout, although medium-severity classes are slightly more challenging than low- and high-severity classes. For instance, the model achieves a classification accuracy of 93%, 88%, 94%, and 90% for cybersickness, cognitive physical load, cognitive mental load, and working memory, respectively. Furthermore, the model achieves an inference time of approximately 1.8 ms per frame, well within the real-time budget for 90 Hz rendering [24].

Reward metric evolution: The PPO training process over 5×10^5 steps reveals three distinct phases in the evolution of reward components across the targeted user cognitive states, as shown in Figure 2. During the initial exploration phase, all reward terms remain negative as the agent tests different visual techniques and intensity levels. In this stage, r_{CS} shows

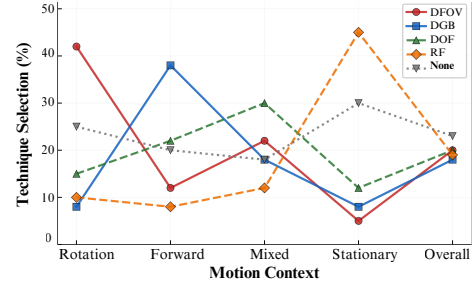


Fig. 3. Distribution of RL-selected visual techniques across different VR motion contexts.

the largest drop because the RL agent often applies stronger techniques to reduce cybersickness, which can temporarily increase cognitive load and reduce working memory (Figure 2a). As training enters the balancing phase, r_{CS} rises sharply once the agent learns more effective combinations of technique, intensity, and motion context, while r_{CPL} and r_{CML} improve more gradually as intensity control becomes more refined. Among the targeted user cognitive states, r_{WM} improves most slowly, consistent with both its lower reward weight ($\alpha_{WM} = 0.10$) and the greater difficulty of preserving working memory while simultaneously reducing cybersickness. In the final convergence phase, all reward terms stabilize with low variance, and the total reward converges near 0.6 with a narrowing ± 1 standard deviation (SD) band, as shown in Figure 2b. This behavior suggests that the penalty terms effectively reduce rapid action switching and promote stable policy behavior. The clear separation between r_{CS} and the remaining reward terms in the converged policy reflects the intended priority structure, where cybersickness reduction receives the strongest emphasis while balancing cognitive load and improving working memory.

Action distribution analysis: We analyze the RL agent's action selections over held-out evaluation episodes using the trained policy described in Section IV. At each decision step, we record the selected visual technique, group actions by motion context (e.g., rotation, forward, mixed, and stationary), and compute the percentage of selections for each technique within each context, as shown in Figure 3. The resulting distribution reveals clear context-dependent behavior. During rotational motion, DFOV is selected most often (42%), consistent with its role in suppressing peripheral optical flow induced by head turns. During forward walking, DGB dominates (38%), reflecting its effectiveness in reducing translational flow during locomotion. In mixed movement contexts, DOF is preferred (30%), suggesting that depth-selective defocusing is better suited to combined rotational and translational motion. During stationary periods, RF peaks at 45%, providing a low-cost visual anchor when cybersickness risk is relatively low. Notably, the None action accounts for 20–30% of selections across contexts, indicating that the agent has learned to withhold visual adaptation when predicted user states remain within acceptable ranges.

VI. USER STUDY DESIGN AND EVALUATION

This section presents the user study design and evaluation results to show the effectiveness of our proposed framework.

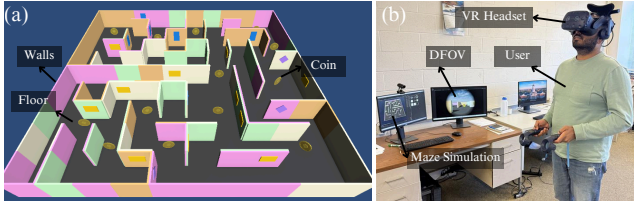


Fig. 4. (a) Top-down view of the procedurally generated VR maze environment. (b) Participant immersed in the proposed AURORA framework.

A. User Study Design on AURORA

1) *Hypotheses*: This study aims to evaluate the effectiveness of AURORA, which uses multi-objective RL to jointly optimize users' cognitive states by dynamically selecting the most suitable visual technique and adaptively adjusting its intensity to reduce cybersickness, balance cognitive load, and improve working memory in VR. Based on this objective, we formulate the following hypotheses to compare AURORA against the conventional method (where the visual techniques are applied using static, fixed, or predefined rules):

- H1**: AURORA with adaptive visual techniques will reduce cybersickness, balance cognitive load, and improve working memory compared to conventional methods.
- H2**: Participants will experience a better UX and a higher sense of presence in the virtual environment when using AURORA compared to the conventional methods.
- H3**: HIL-personalized AURORA will further improve outcomes compared to non-personalized AURORA.

2) *Participants, Study Task, and Procedure*: Our user study involved 47 volunteer participants (26 male, 21 female) aged from 18 to 39 (mean age 27.2, SD 3.26). The participants were Asian, American, Black, and Hispanic. Participants reported no vestibular dysfunction and were not taking anti-motion sickness medication. Thirty-two of the 47 participants reported prior VR experience, whereas the rest stated they had never used VR equipment. Our user study was approved by the Institutional Review Board (IRB) at the authors' university. All participants completed an IRB-approved consent form and a demographic questionnaire (e.g., age, gender, ethnicity, prior VR experience, baseline SSQ, etc.). Participants were briefed on the study procedures and informed of their right to withdraw at any time if they felt too uncomfortable. They then entered the VR study using the HTC Vive Pro Eye headset and controllers and completed a tutorial that outlined the study tasks. Further details are provided below.

Warm-up Phase: Before the experiment, participants were briefed on the study procedures and completed a virtual tutorial introducing the tasks: *coin collection* for cognitive load measurement and *digit span recall* for working memory assessment. They first engaged in a practice session to become familiar with real walking locomotion in the maze and the study tasks. The tutorial also ensured that participants could clearly hear instructions through the HMD's integrated headphones. During the tutorial, participants practiced the *coin-collection task*, where coins were placed at random locations throughout the maze corridors and collected by walking close to them. They also practiced the *digit span recall task*. To assess working memory, we used a VR-based version of the

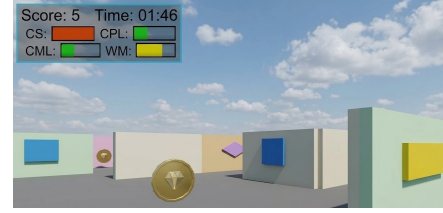


Fig. 5. User's view in the VR maze simulation without any visual technique.

DST [47]. Participants received a randomly generated five-digit sequence every 30 seconds, were given 20 seconds to memorize it, and 10 seconds to recall the digits in the same order during the VR simulation. For example, if given the sequence $\{6 - 2 - 1 - 9 - 5\}$, participants were expected to recall it in the same order. Working memory performance was computed as the percentage of correctly recalled digits, following [4]. Participants were instructed to verbally report their discomfort on an FMS scale from 0 to 10 (0 = no sickness, 10 = severe sickness) whenever a pre-recorded voice prompt was played every 30 seconds during the VR simulation. We used this scale to measure real-time cybersickness and to remain consistent with prior DL-based cybersickness detection studies [28]. Furthermore, we asked a single question adapted from the NASA-TLX once per minute: *How physically and mentally demanding is the task on a scale of 1 (not demanding) to 10 (very demanding)? Please provide two ratings, one for each.* to capture the participants' real-time perception of cognitive physical load and cognitive mental load while navigating the VR environment. We combined both into a single query, as repeatedly querying all user state in detail would itself increase cybersickness, elevate cognitive load, and degrade working memory. Note that these measurement scales are widely adopted in SOTA VR user cognitive state assessment studies [4], [9], [10], [24], [32].

Experiment Phase: After completing the tutorial, the VR maze simulation began and lasted 10 minutes unless participants chose to stop earlier because of discomfort. It is important to note that eye and head positions were carefully calibrated before each simulation. Upon satisfactory calibration, participants were immersed in the VR maze environment. After completing each condition, participants completed the Igroup Presence Questionnaire (IPQ) [48] and were then given a break. They were allowed to rest for as long as needed until they felt ready to continue without discomfort. The experiment concluded once participants had completed all study conditions, including the baseline condition in which no visual technique was applied. To evaluate the effectiveness of the proposed AURORA framework and compare it with conventional methods, the experiment was conducted under the following four conditions:

(1) **Baseline condition**: Initially, the simulation was run without applying any visual technique.

(2) **Static visual techniques**: Next, the visual techniques were applied under static conditions with fixed intensity levels. In this condition, participants experienced all four techniques (DFOV, DGB, DOF, and RF) in a randomized order, one after another within the same condition. Following prior work, the fixed settings were defined as a DFOV aperture of 90° [12],

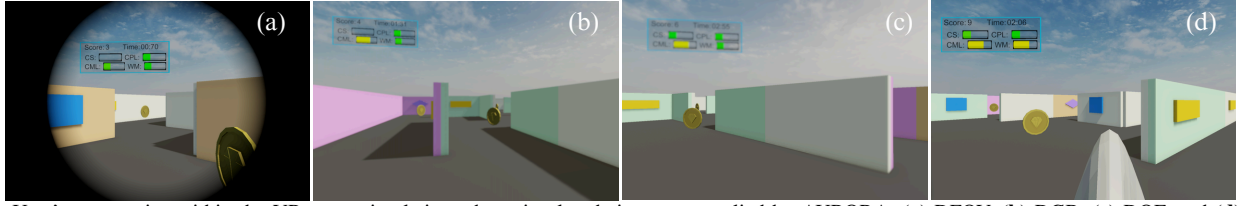


Fig. 6. User's perspective within the VR maze simulation when visual techniques are applied by AURORA: (a) DFOV, (b) DGB, (c) DOF, and (d) RF.

DGB with $\sigma = 8$ [14], DOF with a moderate gaze-contingent focal distance and blur strength [15], and RF with a moderate width and full visibility [17]. Each technique was applied in a fixed, non-adaptive manner to assess its effect on cybersickness, cognitive load, and working memory.

(3) AURORA: Then, as described in Section III-D, the trained multi-objective RL policy was applied, in which the system automatically selected a suitable visual technique from the technique engine and adjusted its intensity in real time.

(4) AURORA-HIL: Finally, as described in Section III-E, the personalized RL agent was evaluated with the same participants on a separate day to assess the effectiveness of policy personalization. For this condition, the RL policy was fine-tuned offline using data collected from earlier conditions, and the resulting personalized policy selected the visual technique and intensity based on each participant's preference profile.

Each condition lasted 10 minutes, but participants were free to stop the experiment at any time if they felt significantly uncomfortable. Figure 4(a) shows the top-down view of the procedurally generated maze, Figure 4(b) shows a participant immersed in the proposed AURORA framework, and Figure 5(a) and Figure 6(a)–(d) illustrate example views of the maze under the baseline and different static visual techniques.

Post-task Interview Phase: In the post-task interview phase, participants first completed the post-SSQ and NASA-TLX. We then conducted semi-structured interviews lasting 1–2 minutes to gather participants' verbal feedback on their experiences, focusing on cybersickness, cognitive load, and working memory across conditions, using a 5-point Likert scale. Afterward, they responded to a set of follow-up statements, such as: “AURORA better balanced my discomfort, cognitive demand, and working memory than the conventional methods”, etc. We also gathered qualitative feedback through open-ended questions on participants' experiences with AURORA compared to conventional methods, beginning with open-ended questions to avoid priming. Before leaving, participants were reminded they could contact the research team if they experienced any discomfort or adverse effects. So far, no concerns have been reported from any participants.

B. User Study Results

This section compares the proposed AURORA framework against conventional methods and the baseline (without visual techniques). We first analyze the experimental results across cybersickness, cognitive load, and working memory during immersion, and then present the post-task evaluation results.

Experiment Phase: The descriptive statistics (e.g., mean (M) and SD) of cybersickness (CS) measured via FMS, cognitive physical load (CPL) and cognitive mental load (CML) measured via the NASA-TLX, and working memory

(WM) measured via the DST across conditions are shown in Table IV. The results show that AURORA (CR) achieves the lowest CS score ($M = 1.52$, $SD = 1.32$), followed by AURORA-HIL ($M = 1.12$, $SD = 1.02$), whereas the baseline produces the highest CS level ($M = 6.30$, $SD = 2.10$). In addition, CR and CR-HIL yield the lowest CPL and CML values and the highest WM values across all conditions, indicating that the proposed method effectively reduces CS while balancing cognitive load and improving WM. To assess the significance of these mean differences, we first apply the Shapiro-Wilk tests, which show that the distributions of CS, CPL, CML, and WM scores deviate from normality in several conditions. We therefore conduct a nonparametric Friedman test, with a significance level of $\alpha = 0.05$. The Friedman test indicates significant differences across conditions for all four user states: CS ($\chi^2(6) = 180.53$, $p < .001$, $W = 0.640$), CPL ($\chi^2(6) = 112.78$, $p < .001$, $W = 0.400$), CML ($\chi^2(6) = 93.54$, $p < .001$, $W = 0.332$), and WM ($\chi^2(6) = 58.88$, $p < .001$, $W = 0.209$), with Kendall's W indicating a large effect for CS, moderate effects for CPL and CML, and a small but meaningful effect for WM. For post hoc comparisons, we conduct Wilcoxon signed-rank tests for each condition pair with Holm-Bonferroni correction, as shown in Table VI. All static visual techniques significantly reduce CS relative to baseline (all $p < .001$), confirming measurable relief from CS. However, their effects on CPL, CML, and WM vary across methods, suggesting that static techniques reduce cybersickness without reliably balancing cognitive load or preserving working memory. In contrast, our proposed AURORA significantly reduces CS compared with all four static visual techniques ($p < .001$) and the baseline ($p < .001$), achieving a 75.8% reduction in CS. At the same time, it improves the other targeted user cognitive states, with lower CPL ($p = .050$, $r = 0.44$), lower CML ($p < .001$), and substantially higher WM ($p = .020$, $r = -0.54$). This pattern differs from static techniques because CR adaptively selects visual techniques and intensities based on the current context and predicted user cognitive states, rather than applying a fixed change throughout the session. As a result, it simultaneously reduces cybersickness while improving cognitive load and working memory across all dimensions. Compared with CR, the static visual techniques also show significantly higher CML for DGB, DOF, and RF (all $p < .001$, $r = 0.64$ – 0.81), higher CPL for DFOV and RF ($p \leq .011$, $r = 0.54$ – 0.62), and lower WM for RF ($p < .001$, $r = -0.68$) and DGB ($p = .037$, $r = -0.50$). These findings support **H1**, confirming that AURORA, with adaptive visual techniques, reduces cybersickness, balances cognitive load, and improves working memory compared with conventional methods.

Beyond AURORA's advantage over conventional methods,

TABLE IV
DESCRIPTIVE STATISTICS (MEAN SD) ACROSS ALL CONDITIONS.

Condition	CS	CPL	CML	WM
Baseline	6.30 (2.10)	5.54 (1.52)	6.15 (1.77)	4.63 (1.90)
DFOV	4.35 (1.78)	5.95 (1.55)	5.09 (1.58)	5.13 (1.78)
DGB	4.80 (1.85)	4.15 (1.28)	6.49 (1.65)	4.75 (1.62)
DOF	3.90 (1.72)	5.05 (1.42)	6.50 (1.55)	4.90 (1.68)
RF	5.50 (2.05)	5.71 (1.48)	6.02 (1.68)	4.33 (1.72)
CR	1.52 (1.32)	4.77 (1.25)	4.33 (1.38)	5.79 (1.45)
CR-HIL	1.12 (1.02)	4.02 (1.18)	3.95 (1.28)	6.02 (1.22)

TABLE V
PERCENTAGE CHANGE (Δ) RELATIVE TO BASELINE. **RED VALUES**
INDICATE WORSENING COMPARED TO BASELINE.

Technique	CS (%)	CPL (%)	CML (%)	WM (%)
DFOV	30.9	7.4	17.2	10.9
DGB	23.8	25.1	5.6	2.7
DOF	38.1	8.8	5.7	5.8
RF	12.7	3.0	2.1	6.5
CR	75.8	13.9	29.6	25.1
CR-HIL	82.2	27.4	35.8	30.0

we further evaluate whether HIL personalization provides additional benefit. A direct comparison between CR and CR-HIL shows descriptive improvements across all four user cognitive states, with CS decreasing from 1.52 to 1.12, CPL from 4.77 to 4.02, CML from 4.33 to 3.95, and WM increasing from 5.79 to 6.02. Relative to the baseline, CR-HIL reduces CS by 82.2%, CPL by 27.4%, and CML by 35.8%, while improving WM by 30.0%, as shown in Table V. Compared with non-personalized AURORA, CS shows a statistically significant improvement ($p = .045$, $r = 0.34$), CPL shows a significant improvement with a large effect ($p = .009$, $r = 0.55$), and WM shows a highly significant improvement ($p < .001$, $r = -0.92$), whereas CML remains stable ($p = .968$, $r = 0.17$). These findings provide strong support for **H3**, suggesting that HIL personalization further reduces cybersickness, balances cognitive load, and improves working memory, consistent with adaptation to individual tolerance profiles [3]. To further evaluate the proposed framework in terms of presence, we compare the IPQ presence scores of CR, CR-HIL, and the conventional methods, collected after each condition on a 7-point Likert scale. The descriptive statistics are as follows: CR ($M = 5.61$, $SD = 1.89$), CR-HIL ($M = 6.13$, $SD = 1.25$), and the average across static techniques ($M = 4.03$, $SD = 2.43$). A Shapiro-Wilk test indicates that the presence scores are not normally distributed. We therefore conduct Mann-Whitney U tests to examine pairwise differences between conditions. The results reveal a significant difference between CR and the static techniques ($Z = 2.73$, $U = 532$, $p = 0.027$), while CR-HIL shows a descriptively higher presence score than both. These findings further support **H2**, suggesting that the proposed framework provides a higher sense of presence and a better overall UX than conventional methods.

Post-task Interview Phase: We also report additional user study results evaluating participants' perceptions of the proposed AURORA framework compared with conventional methods. During the study, participants rated their experiences on a 5-point Likert scale [49], comparing AURORA with conventional methods without RL-based adaptation, where the applied visual techniques remained static, following prior work [9], [24]. Specifically, participants were asked to indicate their agreement with the following statements: **(S1)** AURORA

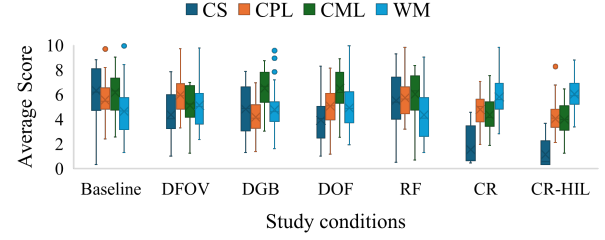


Fig. 7. Average scores (CS,CPL,CML,WM) across study conditions.

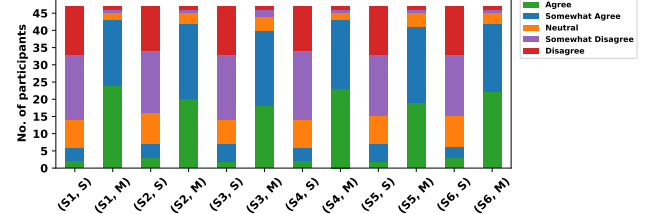


Fig. 8. User feedback in Likert scale between the AURORA framework denoted as (M), and without the AURORA framework denoted as the conventional (S) technique, in which the visual technique is static.

better balanced my discomfort, cognitive demand, and working memory than the conventional methods, (S2) The visual technique selection made by AURORA felt reliable and appropriate based on my VR experience, (S3) The automatic switching between different visual techniques in AURORA felt smooth and non-disruptive, (S4) The personalized version of AURORA felt more tailored to my needs than the non-personalized version, (S5) Using AURORA improved my overall VR experience compared to sessions with conventional methods, and (S6) I would recommend AURORA to others because of its adaptive approach compared to conventional methods. We select these six statements to evaluate the proposed AURORA framework because they collectively capture the key aspects of user experience and system effectiveness needed to assess a new framework, following prior work [9], [24]. To evaluate the rated users' experiences, we compute the mean and SD of the Likert score for each statement across all participants. The user feedback results are shown in Figure 8. We observe that approximately 89% of participants agreed or strongly agreed with all six statements, confirming that AURORA was perceived as more effective at reducing cybersickness, balancing cognitive load, improving working memory, and providing a more immersive experience than conventional methods. At the same time, most participants reported that the conventional methods introduced unnecessary cognitive load and disrupted working memory during immersion. These findings further support **H1**, **H2**, and **H3**, indicating that AURORA is perceived as more effective across the targeted user cognitive states, more immersive, and more beneficial when personalized than conventional methods.

SSQ Results for All Study Conditions: In examining the SSQ scores, we compare the results from the pre-study and the post-study conditions for all participants and find a significant difference. For this, we first perform a Shapiro-Wilk test on these data, indicating that we cannot assume normality. Therefore, we conduct the Wilcoxon signed-rank test to quantify the difference between the pre and post-session SSQ subscores and the total scores. All subscores except Disorientation showed a significant post-session in-

TABLE VI

PAIRWISE WILCOXON SIGNED-RANK COMPARISONS WITH HOLM-BONFERRONI CORRECTION, WHERE CR = AURORA, W = TEST STATISTIC, p (ADJ.) = CORRECTED p -VALUE, r = RANK-BISERIAL EFFECT SIZE.

Condition Pair	FMS			CPL			CML			WM		
	W	p (adj.)	r	W	p (adj.)	r	W	p (adj.)	r	W	p (adj.)	r
Baseline vs. DFOV	34	<.001	0.94	237	0.009	-0.56	106	<.001	0.80	317	0.088	-0.44
Baseline vs. DGB	48	<.001	0.92	13	<.001	0.98	375	0.188	-0.34	496	0.537	-0.12
Baseline vs. DOF	22	<.001	0.96	278	0.017	0.51	368	0.188	-0.35	399	0.488	-0.26
Baseline vs. RF	168	<.001	0.70	387	0.181	-0.31	526	1.000	0.07	375	0.349	0.31
Baseline vs. CR	12	<.001	0.98	315	0.050	0.44	152	<.001	0.73	251	0.020	-0.54
Baseline vs. CR-HIL	5	<.001	0.99	105	<.001	0.81	118	<.001	0.79	226	0.006	-0.60
DFOV vs. DGB	331	0.027	-0.41	0	<.001	1.00	58	<.001	-0.90	319	0.088	0.43
DFOV vs. DOF	278	0.007	0.51	101	<.001	0.82	70	<.001	-0.88	437	0.537	0.23
DFOV vs. RF	104	<.001	-0.82	356	0.111	0.37	199	0.001	-0.65	138	<.001	0.76
DFOV vs. CR	29	<.001	0.95	212	0.002	0.62	341	0.108	0.40	380	0.309	-0.33
DFOV vs. CR-HIL	10	<.001	0.98	48	<.001	0.92	259	0.011	0.54	316	0.112	-0.42
DGB vs. DOF	148	<.001	0.74	80	<.001	-0.86	560	1.000	0.01	451	0.537	-0.20
DGB vs. RF	228	0.001	-0.60	11	<.001	-0.98	323	0.086	0.43	327	0.136	0.40
DGB vs. CR	32	<.001	0.94	333	0.073	-0.41	102	<.001	0.81	284	0.037	-0.50
DGB vs. CR-HIL	9	<.001	0.98	486	0.515	0.14	71	<.001	0.87	234	0.007	-0.59
DOF vs. RF	37	<.001	-0.93	210	0.002	-0.63	328	0.088	0.42	194	0.002	0.66
DOF vs. CR	45	<.001	0.92	457	0.515	0.19	106	<.001	0.81	303	0.062	-0.46
DOF vs. CR-HIL	15	<.001	0.97	191	0.001	0.66	60	<.001	0.89	247	0.011	-0.56
RF vs. CR	13	<.001	0.98	249	0.011	0.54	201	<.001	0.64	180	<.001	-0.68
RF vs. CR-HIL	5	<.001	0.99	64	<.001	0.89	121	<.001	0.79	150	<.001	-0.73
CR vs. CR-HIL	375	0.045	0.34	252	0.009	0.55	450	0.968	0.17	47	<.001	-0.92

crease: Nausea ($Z = -4.12, W = 142, p = 0.003$), Oculomotor ($Z = -3.68, W = 118, p = 0.002$), Disorientation ($Z = -1.92, W = 185, p = 0.076$), and Total Score ($Z = -3.85, W = 105, p = 0.001$).

VII. LIMITATIONS AND FUTURE WORK

The AURORA framework has a few limitations. First, our user study was conducted with a specific participant group in a VR maze navigation task, and we did not explicitly examine how factors such as age, cultural background, or task diversity may influence cybersickness, cognitive load, and working memory [11]. Future work should therefore evaluate AURORA with more diverse participants, broader VR tasks, and longer exposure durations. Second, the MTL-enabled DL model used to predict user states for RL reward shaping is primarily trained on maze-walking data, and the RL policy is also developed in a VR maze simulation. Although domain randomization is applied across varied procedural settings during RL training, the framework has not yet been evaluated beyond this environment. Future work should extend training to multiple environments, evaluate performance on unseen scenarios, and assess AURORA in more diverse VR contexts. Finally, we plan to expand the visual technique library to include additional techniques (e.g., snap turning, color manipulation, etc.) and evaluate the framework across diverse VR environments.

VIII. CONCLUSION

This work proposed AURORA, a multi-objective RL framework that jointly optimized users' cognitive states by reducing cybersickness, balancing cognitive load, and improving working memory capabilities, thereby enhancing UX and comfort during VR immersion. Specifically, the framework incorporates an MTL-enabled DL model that predicts user cognitive states in real time from streaming eye- and head-tracking data, which are then used to construct a multi-objective reward function for RL-based policy learning. Based on this reward, the RL agent dynamically selects visual techniques from our developed visual technique library (DFOV, DGB, DOF, RF) and adaptively adjusts their intensity, enabling context-aware policies that optimize overall UX rather than focusing

solely on cybersickness reduction. Furthermore, a HIL fine-tuning method further personalized the learned policy using in-session comfort ratings and post-session feedback, enabling adaptation to individual tolerance profiles. To validate the framework, we implemented AURORA on a consumer-grade VR headset (HTC Vive Pro Eye), released it as open source, and evaluated it through a user study with 47 participants. Results show that the proposed AURORA significantly reduces cybersickness while balancing cognitive load and improving working memory compared to static techniques, with approximately 89% of participants reporting it as more effective, less demanding, and more immersive. These findings demonstrate that multi-objective RL is a promising approach for jointly optimizing users' cognitive states in consumer VR headsets.

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